

Replicating De Stefano et al. (2025) on Chinese customs data

The firm–product network layer: late-8 sample (2008–2015), HS6

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1 Introduction: what this note does

This note has two purposes, and they map onto the order in which the work was actually done. The first is a *faithful replication*: we take the firm–product network pipeline of De Stefano, Mula, Mariani and Zaccaria (2025, *From macro to micro: economic complexity indicators for firm growth*, arXiv:2507.21754) and re-run it, step by step and with their own formulas, on Chinese firm-level export microdata. The original study works on Italian firms; we ask whether the same machinery — a Balassa-binarised firm–product matrix, Barber’s bipartite modularity (BRIM), in-/out-of-block diversification, Sapling-Similarity coherence and the Fitness–Complexity metric — produces the same structure on a very different economy. It does: we recover the same bipartite modularity ($Q_B = 0.505$ against their 0.501), the same three capability pillars (textiles, chemicals, machinery/metals), and the same core validation that the data-driven blocks are not a relabelling of HS sectors.

The second purpose is to go *beyond* the replication, in the direction of the dynamic, tariff-driven analysis this project is ultimately about. Our microdata carry information the Italian data did not foreground — the export *destination* — and so, once the static blocks are in place, we ask *which firms* actually diversify outside their capability block, and *which blocks* those firms sit in. The answer turns out to be sharp and economically meaningful, and it tells us a good deal about how to set up the panel: out-of-block exploration is the signature of *geographic breadth* (serving many destinations), not of export volume, and it is concentrated in particular blocks and firm types. The final section collects what all of this implies for the dynamic landscape we will build next.

Throughout we are explicit about what is and is not reproducible on these data: EXPY/PRODY and the growth and profit regressions require world trade data and firm financials we do not have, and we flag them as documented, not faked.

2 Replication of the De Stefano pipeline

2.1 The pipeline and its fidelity

De Stefano build firm-level economic-complexity indicators from a bipartite firm–product export network. Their pipeline, which we follow in order, keeps only firms exporting at least one product *every year* (a balanced panel), aggregates exports into a single matrix and binarises it with the Balassa Revealed Comparative Advantage; detects capability **blocks** with BRIM (Barber, 2007); defines **in-block** and **out-of-block diversification** $d^{\text{in}}, d^{\text{out}}$ from $\delta_{fp} = \mathbb{1}\{\text{block}(f) = \text{block}(p)\}$; measures portfolio **coherence** with the Sapling-Similarity product kernel (Albora et al., 2023); computes **Fitness–Complexity** (Tacchella et al., 2012); and validates the partition by showing the blocks carry information beyond naive HS-sector dummies (their Table 2).

Table 1 is the honest audit of how closely we match them. The core network pipeline is reproduced faithfully; the one deliberate departure is the time window — an eight-year sub-period rather than their twenty-five years — together with two minor approximations and one addition (we also show Louvain and Leiden alongside BRIM, and we add the destination analysis of §3).

Table 1: Fidelity audit: our pipeline vs. De Stefano et al. (2025).

Step	De Stefano	This work	Identical?
Firm–product RCA	Balassa (their Eq. 1)	same Balassa formula	yes
Threshold	$\text{RCA} > 1$ (strict)	$\text{RCA} > 1$ (strict)	yes
Aggregation	average connections over all years	sum export value over years	equivalent ^a
Sample rule	≥ 1 product every year	≥ 1 product every year	yes
Sample window	1993–2017 (25 yr)	2008–2015 (late-8, 8 yr)	no (by design)
Community detection	BRIM only	BRIM + Louvain/Leiden shown	superset
Coherence	all products	top-1500 products (compute cap)	approximation
$d^{\text{in}}/d^{\text{out}}, \delta_{fp}$	as defined	identical	yes
EXPY/PRODY, regressions	computed	documented, not computed	data-limited

^a Under a balanced panel, $\text{mean} = \text{sum}/T$ with T constant; RCA is scale-invariant to a global constant, so the two yield a bit-identical binary matrix (verified: 0 differing cells).

2.2 Data, sample, and the bipartite network

The raw input is Chinese firm $\times$ product $\times$ destination $\times$ year exports. For the late-8 window (2008–2015) the firm–product–year panel has 11,923,680 rows over 372,030 firms and 4,657 HS6 products. Applying De Stefano’s balanced rule — exports in *every* year of the window — gives the working sample in Table 2.

Table 2: Sample coverage under the balanced (“every year”) rule, late-8 / HS6.

Quantity	Value
Firms, all (any year in window)	372,030
Firms, balanced (every year)	71,017
share kept	19.1%
HS6 products (balanced sample)	4,586
Export value share retained	62.1%
Mean HS6 lines per firm-year	9.5

The balanced rule is demanding — only 19% of firms survive it — but those firms carry 62% of export value, so the panel is small in count yet economically dominant: it is the established-exporter core, not the full population. That selection is worth keeping in mind for external validity, because the rule discards the entry- and exit-heavy tail that is so characteristic of Chinese customs data. We return to this when we discuss the dynamic extension.

The bipartite network. With pooled export value X_{fp} , the Balassa index and its binarisation are

$$\text{RCA}_{fp} = \frac{X_{fp} / \sum_{p'} X_{fp'}}{\sum_{f'} X_{f'p} / \sum_{f'p'} X_{f'p'}}, \quad M_{fp} = \mathbb{1}\{\text{RCA}_{fp} > 1\}. \quad (1)$$

The resulting incidence has 71,017 active firms, 4,586 used products and 1,103,737 links. Firm *diversification* $d_f = \sum_p M_{fp}$ has mean 15.5, median 8 and a maximum of 1,005, with 6% of firms single-product; product *ubiquity* $u_p = \sum_f M_{fp}$ has mean 241, median 101. Both distributions are heavy-tailed (Fig. 1), the usual economic-complexity signature: most firms are specialised and a few hugely diversified, most products niche and a few near-universal.

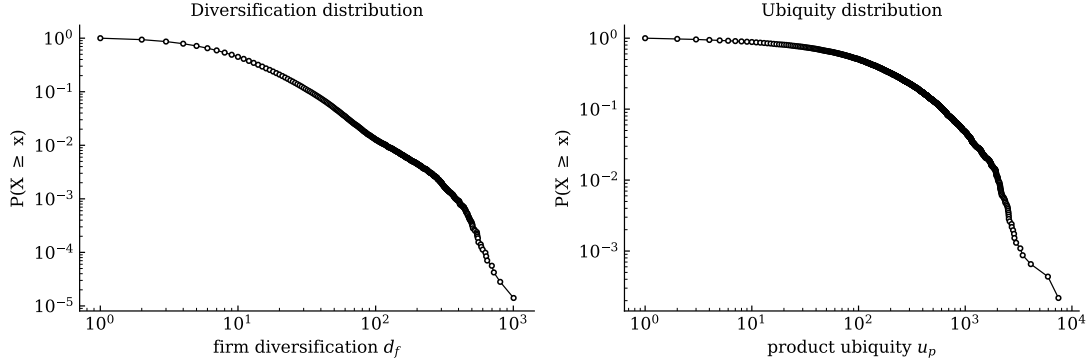


Figure 1: Complementary CDFs of firm diversification d_f (left) and product ubiquity u_p (right), on log-log axes.

2.3 Community detection

Standard modularity (Newman, 2006) and its bipartite (Barber, 2007) counterpart for a firm×product incidence B , with firm degrees k , product degrees d and $m = \sum_{fp} B_{fp}$, are

$$Q = \frac{1}{2m} \sum_{ij} \left(A_{ij} - \frac{k_i k_j}{2m} \right) \delta(c_i, c_j), \quad Q_B = \frac{1}{m} \sum_{fp} \left(B_{fp} - \frac{k_f d_p}{m} \right) \delta(g_f, g_p). \quad (2)$$

Louvain and Leiden optimise the standard Q on the bipartite graph; BRIM optimises Q_B , the correct bipartite null, and is De Stefano’s choice, which we carry forward as canonical. Table 3 compares the three.

Table 3: Community detection on the same incidence (late-8, HS6). All at resolution $\gamma = 1$.

Algorithm	#blocks	Q_B	Agreement (firm side)	ARI	NMI
Louvain	13	0.521	Louvain vs Leiden	0.558	0.669
Leiden	9	0.538	Louvain vs BRIM	0.425	0.584
BRIM (canonical)	5	0.505	Leiden vs BRIM	0.455	0.615
<i>De Stefano (Italy): 7 blocks, $Q_B = 0.501$</i>					

The headline is that our BRIM modularity, $Q_B = 0.505$, is essentially identical to their 0.501: the *strength* of the modular structure replicates almost exactly across two very different economies. The *number* of blocks, by contrast, is not an invariant — it depends on the algorithm (Louvain and Leiden over-segment bipartite graphs) and on the resolution γ (on this same network the count runs $5 \rightarrow 10 \rightarrow 24 \rightarrow 70$ as $\gamma = 1 \rightarrow 2 \rightarrow 4 \rightarrow 8$). The comparison with their seven is therefore meaningful only at the same $\gamma = 1$ and the same (BRIM) algorithm, a point worth remembering before reading anything into “5 vs 7”.

What γ is. The resolution γ weights the null term in the modularity. Each pair of nodes placed in the same block is “charged” γ times its expected random-baseline connection $k_f d_p / m$, against a reward of +1 per observed link; a block survives only when its internal reward beats that charge. So $\gamma = 1$ is standard modularity — the charge equals the configuration-model expectation, De Stefano’s choice — while $\gamma > 1$ raises the charge and yields more, smaller blocks, and $\gamma < 1$ yields fewer, larger ones. In the CPM-bipartite implementation we use, γ is equivalently a *density threshold*: a community is kept only if its internal link density exceeds γ . The Leiden routine is only the optimiser; γ is a property of the objective, not of the algorithm. The Appendix details the three algorithms and their null models. We keep $\gamma = 1$ throughout and report $\gamma = 2$ as a robustness check (§4.1).

2.4 Block anatomy and the De Stefano comparison

2.4.1 What our blocks contain

Table 4 reports the five BRIM blocks with their dominant HS chapters and representative products.

Table 4: The five BRIM blocks (late-8, HS6; 71,017 firms).

Block	Firms	Dominant content
0	36,321 (51%)	Machinery & electrical 36%, base metals 24%, instruments 7% — the large <i>generalist</i> block (absorbs electrical machinery)
2	16,095 (23%)	Paper 16%, misc. manufacturing 12%, packaging/cases — light manufacturing
1	10,870 (15%)	Textiles 95% — apparel, cotton, man-made fibres (pure)
3	7,691 (11%)	Chemicals 48% + agri-food ~41% (vegetables 18%, foodstuffs 12%, animals 11%)
4	40 (0%)	Precious 100% — diamonds (pure satellite)

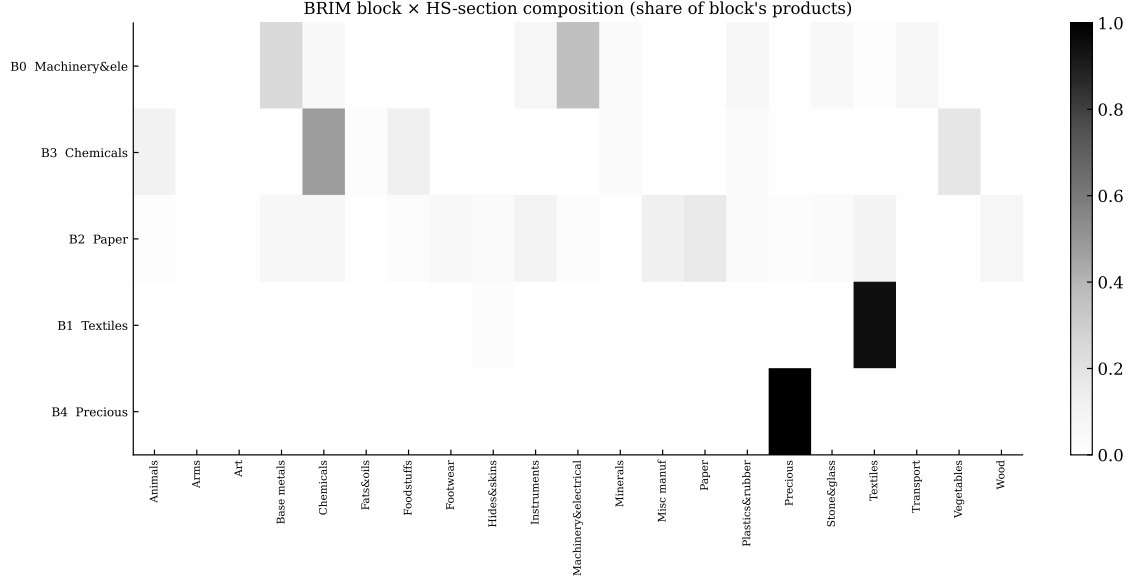


Figure 2: BRIM block $\times$ HS-section composition (share of each block’s products in each section). Textiles (block 1) is near-pure; the generalist block 0 spreads across machinery, metals and instruments.

2.4.2 Comparison with De Stefano’s seven blocks

The mapping in Table 5 shows the differences are economically sensible rather than failures of replication.

Table 5: Block-by-block comparison (validation).

De Stefano (Italy, 7)	Ours (China, 5)	Match
Machinery & metal (largest, ~30%)	Block 0 (~51%): machinery, metals + <i>electrical</i>	✓ larger, absorbs electrical
Electrical machinery (TVs, hard disks)	folded into Block 0	× does not separate
Chemicals & minerals	Block 3 (chemicals 48% + agri-food)	✓ chemicals; food bundled in
Paper & plastic	Block 2 (paper, packaging, light mfg)	✓ partial
Textile products	Block 1 (cotton, apparel)	✓ strong, pure
Food & animals	bundled into Block 3	~ no <i>standalone</i> food block
Wood/glass/jewelry (least interpretable)	Block 4 (diamonds, 40 firms)	~ partial

Three economically robust pillars — textiles, chemicals, machinery/metals — emerge in both countries with coherent content and near-identical Q_B , so the block structure is not a country artefact. Where China differs, it differs for a reason: machinery, metals and electronics collapse into one large generalist block (~51% of firms, the “factory of the world” assembly base), and agri-food does not nucleate its own cluster but bundles with chemicals. This yields five blocks rather than seven — a *merging* difference, at the same modularity.

2.4.3 The De Stefano robustness check

We cannot run their growth regression (we have no firm outcomes), so we test the same claim at the partition level: the BRIM product-partition against the HS-section partition gives $ARI = 0.289$ and $NMI = 0.426$, with blocks spanning on average thirteen HS sections (min 1, max 21).

The data-driven partition is therefore not a relabelling of HS sectors — exactly their Table 2 point, reproduced.

2.5 In-block and out-of-block diversification

With $\delta_{fp} = \mathbb{1}\{\text{block}(f) = \text{block}(p)\}$, the diversification split is

$$d_f^{\text{in}} = \sum_p M_{fp} \delta_{fp}, \quad d_f^{\text{out}} = \sum_p M_{fp} (1 - \delta_{fp}), \quad d_f = d_f^{\text{in}} + d_f^{\text{out}}, \quad \text{Share}_f^{\text{in}} = \frac{d_f^{\text{in}}}{d_f}. \quad (3)$$

In words, d^{in} counts a firm’s competitive ($\text{RCA} > 1$) products that lie *inside* its own capability block, d^{out} those *outside* it, and the share is the proportion staying in-block. Table 6 reports the distribution.

Table 6: Diversification indicators, late-8 / HS6 ($N = 71,017$ active firms).

	mean	sd	p10	p25	median	p75	p90
d^{in}	13.01	20.83	2	4	7	14	29
d^{out}	2.53	11.38	0	0	0	2	5
d	15.54	28.53	2	4	8	17	33
Share^{in}	0.888	0.162	0.63	0.80	1.00	1.00	1.00

Pure in-block ($\text{Share}^{\text{in}} = 1$): 54.5%; pure out-of-block: 0%; mixed: 45.5%.

The picture is one of deep specialisation. In-block diversification dominates; d^{out} is zero for 54.5% of firms (equivalently, $\text{Share}^{\text{in}} = 1$ for that majority), and where it is positive it is small. Reading the right panel of Fig. 3, firms grow along the d^{in} axis, within their block, which is the exploitation–exploration language of March (1991): these established Chinese exporters are exploitation-heavy. That zero-inflation and that ceiling at $\text{Share}^{\text{in}} = 1$ are not just descriptive curiosities — they are the central fact a dynamic model of these outcomes will have to confront, and we come back to them in the final section.

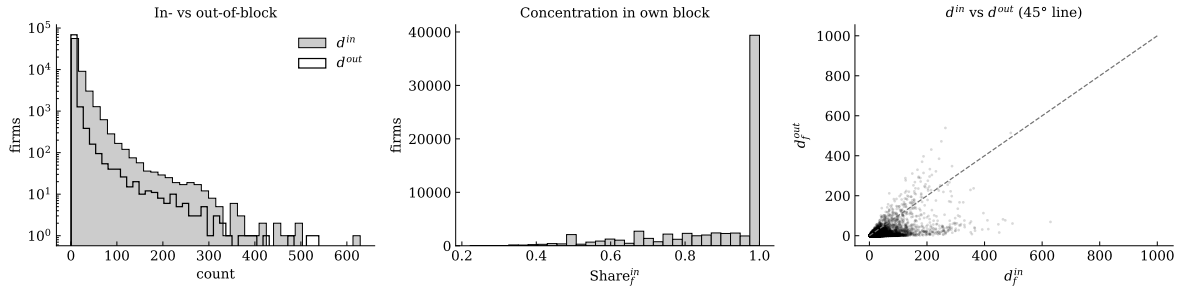


Figure 3: **Left:** in- vs out-of-block diversification (log y). d^{in} dominates; d^{out} is zero for 54.5% of firms (the median firm has $d^{\text{out}} = 0$). **Centre:** concentration Share^{in} — a mass of 54.5% at 1 (pure specialists, equivalently $d^{\text{out}} = 0$). **Right:** d^{in} vs d^{out} with the 45° line; the cloud lies below it, firms grow along the d^{in} axis.

2.6 Coherence and Fitness

Coherence (Sapling Similarity). $C_f = \sum_{p,p'} E_{fp} E_{fp'} B_{pp'} / (\sum_p E_{fp})^2$ with the Sapling kernel B (Albora et al., 2023), computed for 68,434 firms (top-1500 products), with mean 0.626 and range $[0.02, 1.00]$. As one would expect, more diversified firms have somewhat *lower* coherence (Fig. 4): broader baskets are less internally related.

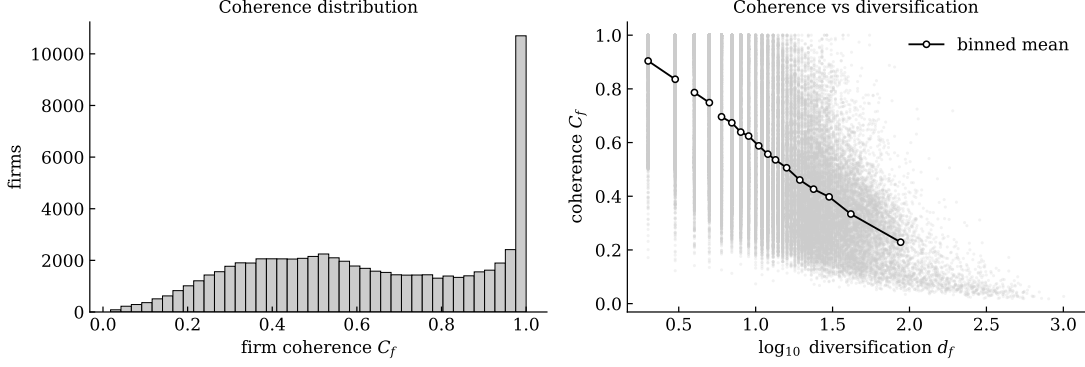


Figure 4: Distribution of firm coherence C_f (left) and coherence vs. diversification with binned mean (right).

For De Stefano coherence is a descriptive indicator; for us it will do identifying work in the dynamic extension, and it is worth saying why now. **Export intermediaries** — trading companies and wholesalers — carry baskets of *unrelated* products, shipping whatever crosses their desk rather than a capability-driven portfolio. They therefore display artificially high d^{out} and very *low* coherence. Left uncontrolled, they would masquerade as “explorers” and contaminate the out-of-block signal precisely where the dynamic story lives. Coherence C_f flags them: conditioning on it, or trimming the low-coherence tail, separates genuine capability reconfiguration from intermediary churn. That is why we compute it now, even though it is not part of the headline replication.

Fitness–Complexity (Tacchella et al., 2012). The non-linear map, iterated to a fixed point, is

$$F_f^{(n)} = \sum_p M_{fp} Q_p^{(n-1)}, \quad Q_p^{(n)} = \frac{1}{\sum_f M_{fp} / F_f^{(n-1)}}. \quad (4)$$

We reproduce the indicator (Fig. 5); fitness rises with diversification but not one-for-one. We cannot test their Appendix-C finding that complexity does not predict growth, since that needs firm outcomes.



Figure 5: Firm fitness vs. diversification (left, binned mean) and mean product complexity by HS section (right).

What we cannot replicate. Two parts of De Stefano’s analysis require data outside a single country’s firm exports, and we flag them honestly rather than approximate them. EXPY and PRODY (Hausmann–Hwang–Rodrik, 2007) are built from the *world* country–product matrix and GDP per capita; a single country’s firm exports give no cross-country variation to estimate PRODY. The growth and profit-per-employee regressions need ORBIS firm financials, which

we do not have. Everything else — RCA, BRIM, $d^{\text{in}}/d^{\text{out}}$, coherence, fitness — is computed exactly as they describe.

3 Extension I — Who explores out of block? Destinations and the multinational reading

De Stefano stop at the firm–product layer. Our microdata also carry the export *destination*, which lets us give d^{out} an economic identity, and ask the question that matters for what comes next: are the out-of-block explorers the wide-reaching, “multinational-like” exporters? For each firm we measure *destination reach* n_f^{dest} (the number of distinct destination countries served over 2008–2015) and *flow* (total export value), and cut the 71,017 firms into 5×5 quintiles of flow $\times$ reach.

The headline is a contrast of two correlations. Out-of-block diversification correlates with destination reach at 0.18, but with export value at only 0.03 (Spearman). Reaching more markets tracks out-of-block exploration; sheer revenue does not. Table 7 makes the pattern transparent on the size-free out-of-block share: reading *across* a row, toward more destinations, the share rises monotonically at every flow level; reading *down* a column, toward more volume, it falls. Importantly, along the destination axis the absolute count d^{out} and the share move together — both rise with reach in every flow row — so destinations are unambiguous. The count-versus-share tension appears only on the flow axis, where the count rises but the share falls; we explain that next.

Table 7: Out-of-block share % by export-flow quintile (rows, F1 low $\rightarrow$ F5 high) $\times$ destination-reach quintile (columns), late-8 / HS6. Median destinations per column: D1 $\approx$ 3, D2 $\approx$ 9, D3 $\approx$ 17, D4 $\approx$ 30, D5 $\approx$ 54.

Export flow	D1	D2	D3	D4	D5
F1 (low)	10.0	11.5	13.1	14.0	14.8
F2	8.7	10.5	12.1	13.9	16.3
F3	7.7	8.8	10.5	13.0	15.2
F4	7.6	8.1	9.6	11.5	14.6
F5 (high)	7.4	7.3	8.5	9.6	13.7

Across a row (more destinations) the share rises; down a column (more volume) it falls. Breadth tilts the portfolio outward; volume tilts it inward.

The raw count tells a complementary story (Fig. 6, left, log scale): the flow lines overlap across the first four destination quintiles and fan out only in the last, where the top-flow line jumps from $d^{\text{out}} \approx 1.05$ in the low-flow/few-destination corner to 7.3 in the high-flow/many-destination corner, a factor of seven. But the difference-in-differences across the corners is +2.9 products on the count and only +1.5 pp on the size-free share, so the apparent “synergy” of being both large and broad is largely the big-and-broad firms simply being large. On the proportion the flow lines stay roughly parallel, with no strong size-free complementarity.

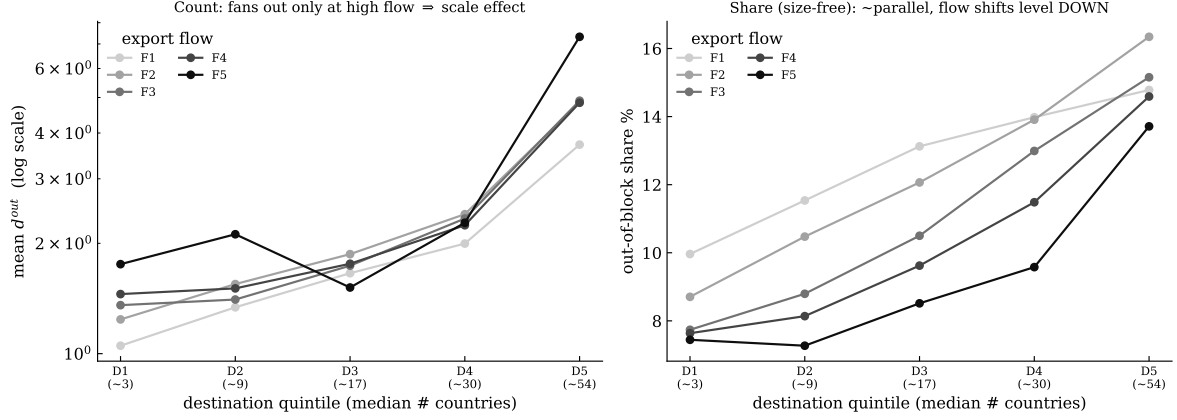


Figure 6: Out-of-block exploration by export flow $\times$ destination reach (five flow quintiles, light = low to dark = high flow). **Left:** mean d^{out} count (log scale) — every line rises with destinations; the lines additionally fan out at the highest destination quintile (a scale effect). **Right:** size-free out-of-block share — every flow line again rises with destinations, and the high-flow lines sit *below*. Both panels rise along the destination axis; they part only on the flow axis.

That volume should *lower* the share looks paradoxical — shouldn't bigger firms do more of everything? They do, in absolute terms; the subtlety is on the proportion. Holding destination reach fixed, going from the lowest to the highest flow quintile raises both margins ($d^{in} \times 1.86$, $d^{out} \times 1.53$), but d^{in} grows more in proportion, so the out-of-block share falls from 12.7% to 9.3% (Fig. 7). The reason is that RCA is scale-invariant: a high-value firm is not just “the same firm, bigger” but one with a more core-concentrated product mix. Scaling in export value comes from deepening in the core — specialisation — not from scattering into unrelated products. Geographic breadth and export volume are therefore different, opposing margins: the extensive margin of markets pushes the portfolio out of block, the intensive margin of value pulls it in. This mechanism is, however, the *least robust* of our findings: it is clean and monotone at $\gamma = 1$ but partially reverses at the highest flow quintile under the finer $\gamma = 2$ partition (§4.1), so we read it as a $\gamma = 1$ pattern rather than a law. The destination result above does not have this fragility — it is monotone at both resolutions.

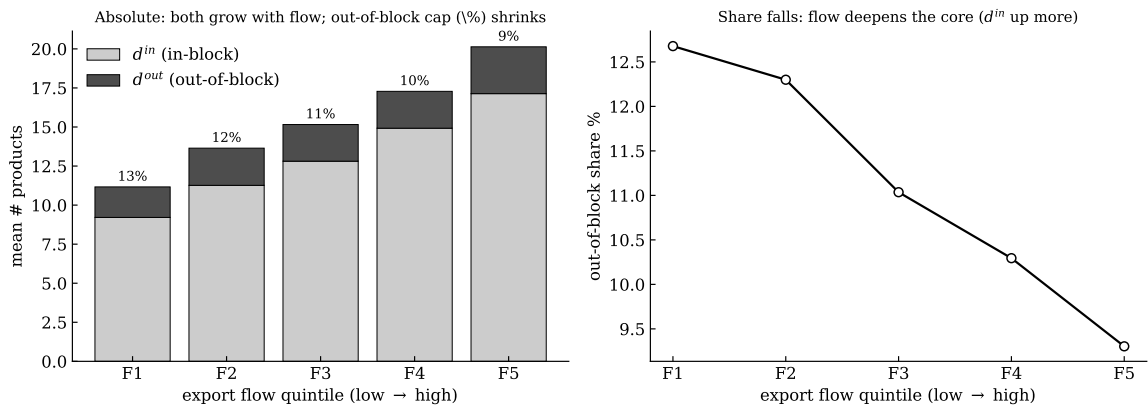


Figure 7: Flow mechanism, destination reach held fixed (equal-weight average over destination quintiles). **Left:** stacked $d^{in} + d^{out}$ by flow quintile — the basket grows but the out-of-block cap thins from 13% to 9%. **Right:** the out-of-block share falls with flow: scaling in value deepens the core.

Economically this is again the exploration–exploitation logic of March (1991). Exploring outside one’s capability block is costly, and it is the firms exposed to many heterogeneous markets — and with the organisational slack to act on that exposure — that do it, not the

firms that simply pump volume through their core. So d^{out} is the signature of the wide-reach exporter, not the big-revenue one. The “multinational-like” type, if we want to talk about one, should be defined on geographic breadth; defining it on volume is not what the data support.

4 Extension II — Where d^{out} lives: across the capability blocks

Having seen *which firms* explore, we ask *which blocks* they sit in. Table 8 distributes d^{in} , d^{out} and the out-of-block share across the five BRIM blocks, with each block’s median reach, median export and product count. One structural caveat is needed to read it: the out-of-block share depends mechanically on block size, because a firm in a large block has a big in-block menu and so few of its products fall outside, whereas a firm in a small, pure block has few in-block options and a larger share out by construction.

Table 8: In-/out-of-block diversification by BRIM block (sorted by out-of-block share). Export in raw USD.

Block	firms	d^{in}	d^{out}	share ^{out}	med. dest	med. exp.	#prod.
B2 Paper / light-mfg	16,095	8.56	4.46	21.5%	22	\$12.0M	715
B3 Chemicals + agri-food	7,691	9.01	1.50	9.0%	14	\$19.9M	1,173
B1 Textiles	10,870	24.55	3.10	8.6%	15	\$21.0M	698
B0 Machinery & electr.	36,321	12.39	1.73	7.9%	17	\$14.9M	1,995
B4 Precious (diamonds)	40	1.55	0.02	0.6%	3	\$57.8M	5

Even with that caveat, the blocks sort into recognisable firm types. **Paper / light manufacturing (B2)** is the explorer block by a wide margin: an out-of-block share of 21.5% and $d^{\text{out}} = 4.46$, roughly three times every other block, on firms that also reach the most markets (median 22 destinations). **Textiles (B1)** is the opposite — a d^{in} of 24.6, the highest by far, yet an out-of-block share of only 8.6%: textile firms diversify enormously, but *within* textiles (yarns, fabrics, the many apparel lines), not outside. **Precious / diamonds (B4)** is the limiting case of the volume story from §3: the highest median export of all (\$57.8M) but only three destinations and essentially no out-of-block activity — a few large, ultra-specialised intensive-margin traders. **Machinery (B0)** and **chemicals (B3)** are the big generalist blocks, with structurally lower out-of-block shares because their in-block menus (1,995 and 1,173 products) are so large.

It is worth pausing on the contrast between Textiles and Machinery, because at first glance it looks contradictory: Textiles has a *higher* d^{in} (24.6 vs 12.4) despite a *smaller* block (698 vs 1,995 products). The resolution is that block size is an aggregate property — how many distinct products exist in the block across all firms — while d^{in} is per firm. A textile firm covers about 3.5% of its block’s products (24.6/698); a machinery firm only 0.6% (12.4/1,995). Textiles is a narrow, deep block of closely related lines that each firm makes many of; machinery is a wide, shallow galaxy of heterogeneous niches, large in total but sparse per firm.

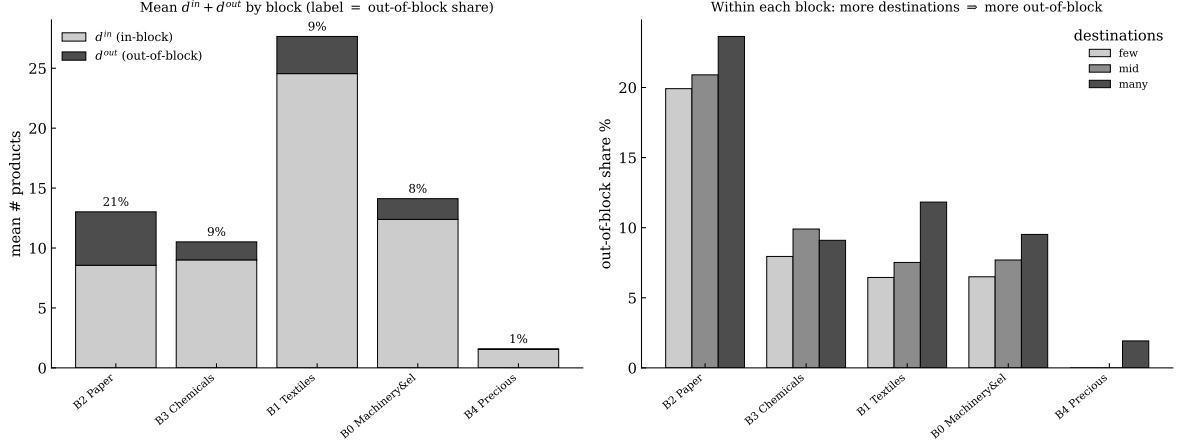


Figure 8: Left: mean $d^{\text{in}} + d^{\text{out}}$ by block (label = out-of-block share); the out-of-block cap is by far thickest for Paper. **Right:** out-of-block share by within-block destination tercile (few/mid/many). The “more destinations $\Rightarrow$ more out-of-block” pattern holds inside every block, so it is not an artefact of composition across blocks.

The right panel of Fig. 8 adds a reassuring robustness check: the destination gradient of §3 reappears *within* each block — more destinations means a higher out-of-block share in Paper, Textiles, Machinery and (more weakly) Chemicals alike. The link between geographic breadth and out-of-block exploration is therefore not a between-block composition effect; it operates inside each capability community.

4.1 Resolution robustness: $\gamma = 1$ vs $\gamma = 2$

The block *count* depends on the resolution parameter γ , which weights the bipartite null in the modularity. De Stefano use standard modularity, $\gamma = 1$, which we keep as the baseline of the whole note; to check that the substantive findings do not hinge on it, we re-ran the *entire* pipeline at $\gamma = 2$ — a finer resolution that penalises the null more, and so yields more, smaller blocks. Table 9 puts the two side by side.

Table 9: The same pipeline at two resolutions. Levels rescale; conclusions do not.

Quantity	$\gamma = 1$ (baseline)	$\gamma = 2$
BRIM blocks	5	10
Bipartite modularity Q_B	0.505	0.544
ARI vs HS sections	0.289	0.318
NMI vs HS sections	0.426	0.476
HS sections spanned per block (avg)	13	9.4
Block purity (mean dominant chapter)	59%	66%
Pure in-block ($\text{Share}^{\text{in}} = 1$)	54.5%	43.2%
$\text{corr}(d^{\text{out}}, \text{reach})$	0.18	0.16
$\text{corr}(d^{\text{out}}, \text{value})$	0.03	0.06

Doubling the resolution splits the five blocks into ten and raises modularity (Q_B : 0.505 $\rightarrow$ 0.544), as expected. The split is partly meaningful and partly mechanical. **Meaningful:** agri-food, bundled into the chemicals block at $\gamma = 1$, separates into its own block at $\gamma = 2$ (vegetables/animals/foodstuffs), and dedicated footwear and transport blocks emerge; HS-sections-per-block falls from 13 to 9.4 and per-block purity rises. **Mechanical:** the coherent textile block splits into two near-identical pure-textile sub-blocks (over-segmentation), and the machinery–metals core stays mixed at any resolution — firm-weighted purity barely moves

(42% $\rightarrow$ 45%), because the bulk of firms sit in that entangled generalist block. Table 10 traces how each $\gamma = 1$ block maps to its $\gamma = 2$ descendants, and Fig. 9 shows the ten finer blocks against HS sections (to be read against Fig. 2).

Table 10: How the five $\gamma = 1$ blocks split at $\gamma = 2$.

$\gamma = 1$ block	$\rightarrow \gamma = 2$ descendants	nature
B0 machinery generalist (51%)	machinery+metals (30%), ery+instruments (15%), transport	core stays mixed; instruments/transport peel off
B3 chemicals + agri-food (11%)	chemicals 65%, <i>agri-food</i> (own block)	genuine separation
B1 textiles 95% (15%)	two pure-textile blocks (90%, 97%)	over-split (artefact)
B2 paper/light-mfg catch-all (23%)	catch-all (27%), footwear	footwear peels off
B4 precious (0%)	precious (unchanged)	unchanged

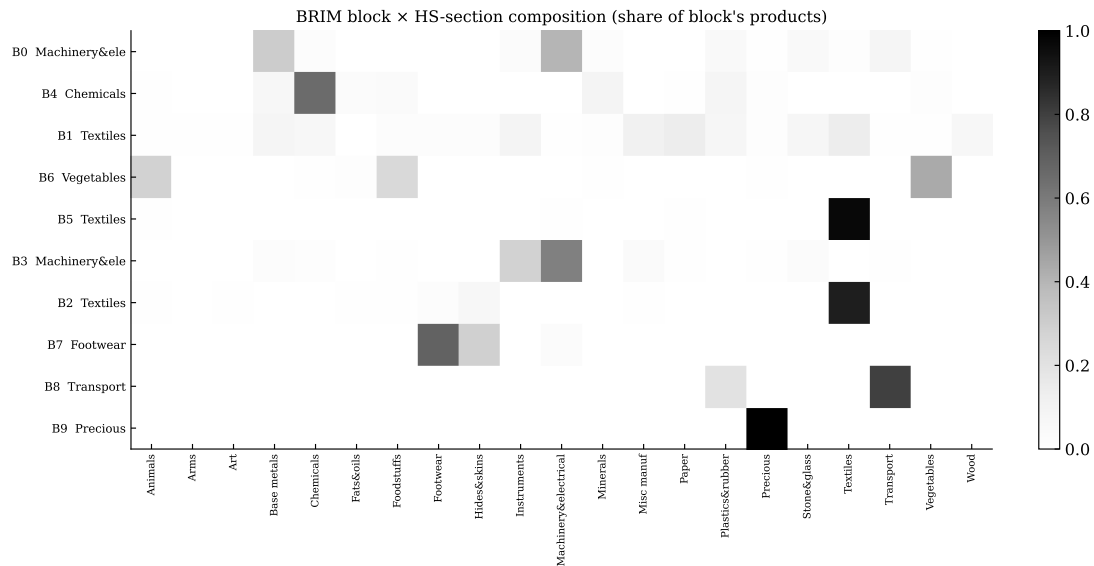


Figure 9: $\gamma = 2$: BRIM block $\times$ HS-section composition (ten blocks). Pure blocks show as isolated dark cells — textiles (two), chemicals, footwear, transport, precious — while block 0 (machinery+metals) and the catch-all block 1 stay spread. Compare with the five-block $\gamma = 1$ map, Fig. 2.

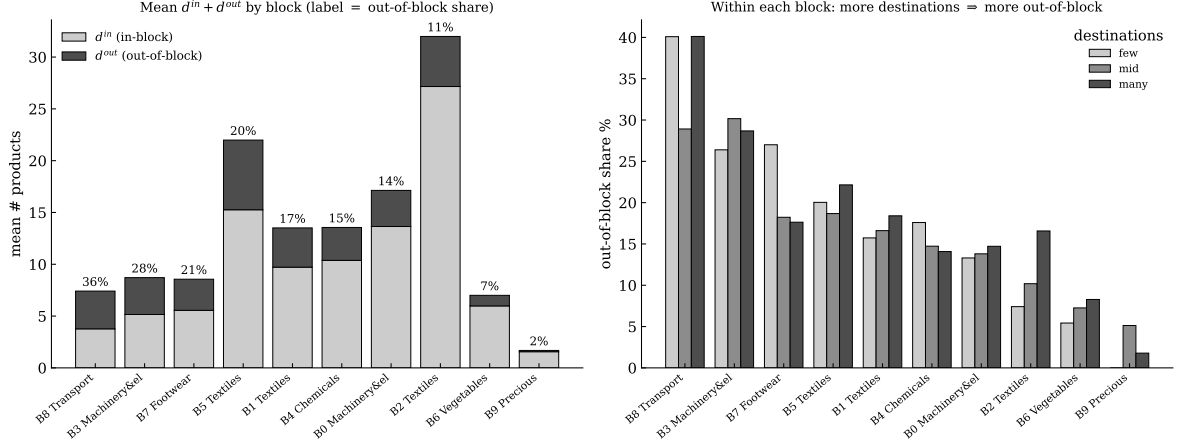


Figure 10: $\gamma = 2$: in-/out-of-block diversification by block (left) and the within-block destination gradient (right). As at $\gamma = 1$ (Fig. 8), out-of-block exploration still rises with destination reach inside the blocks; the absolute shares are uniformly higher because the finer blocks have smaller in-block menus.

The conclusions survive; only the levels rescale. Finer blocks mean a smaller in-block menu, so the out-of-block share rises across the board (pure in-block 54.5% $\rightarrow$ 43.2%) and is *not* comparable across γ — read d^{out} levels within a resolution, never across. But every substantive conclusion holds at $\gamma = 2$: the modular structure is real and in fact stronger (Q_B higher); the blocks are still not a relabelling of HS sectors (ARI/NMI rise only marginally, to 0.32/0.48, and stay far below 1 — the finer blocks are slightly more sector-like, as one would expect, but nowhere near a relabelling); and out-of-block exploration still tracks destination breadth, not export volume (correlation with reach 0.16 against 0.06 with value). The one result that does *not* survive cleanly is the flow mechanism. At $\gamma = 1$ the out-of-block share falls monotonically with export volume (12.7% $\rightarrow$ 9.3%, destination reach held fixed); at $\gamma = 2$ it falls only through the fourth flow quintile and then *rebounds* at the top quintile (15.6% $\rightarrow$ 16.9%), because there out-of-block diversification grows faster than in-block ($\times 1.84$ vs $\times 1.79$). Fig. 11 shows the reversal. We therefore treat “volume deepens the core” as a $\gamma = 1$ pattern, not a robust law — unlike the destination result, which is monotone at both resolutions. Overall the reading is still reassuring: $\gamma = 1$ is a sensible reporting default, giving five economically interpretable blocks without over-splitting coherent sectors, and the substantive findings do not depend on that choice.

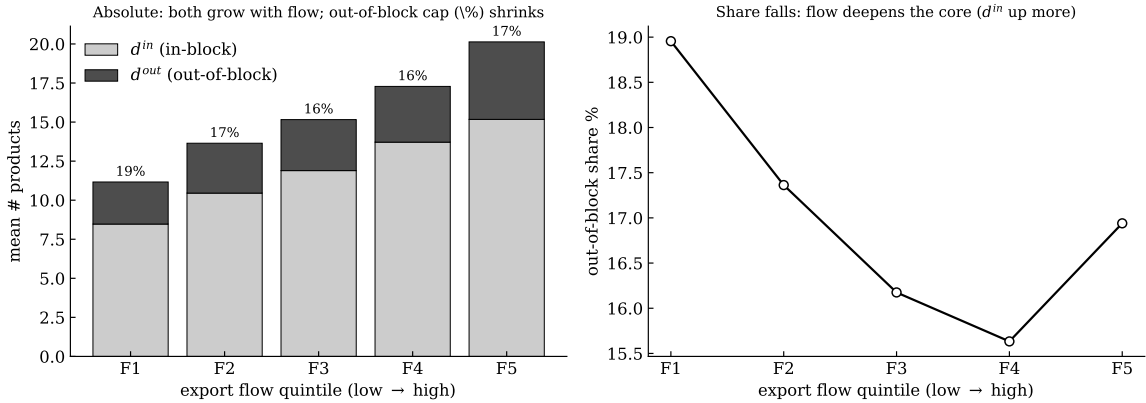


Figure 11: $\gamma = 2$ flow mechanism (destination reach held fixed), to be read against the $\gamma = 1$ version (Fig. 7). **Right:** the out-of-block share falls with flow only up to F4 and then *rebounds* at the top flow quintile F5 — the one place the “volume deepens the core” reading breaks under a finer partition.

5 Keep attention for the dynamic landscape

These are the emphasis points from the static analysis worth keeping in mind for the dynamic, tariff-driven part:

- **Freeze the partition.** Estimate the blocks once (pre-period or pooled) and hold them fixed; otherwise $d^{\text{in}}/d^{\text{out}}$ become endogenous to the shock and the treatment is absorbed into the partition. Let only the portfolio $E_{fp}(t)$ vary: $d^{\text{in}}(t) = \sum_p E_{fp}(t) \delta_{fp}$.
- **Measure within-firm variation first.** A fixed-effects panel identifies off within-firm movement; the year-to-year variance of $d^{\text{in}}, d^{\text{out}}$, share is not yet known and decides power.
- **Pick outcomes by their distribution.** d^{in} is cleanest; d^{out} is zero-inflated (54.5% zeros) $\rightarrow$ PPML; the share is floor/ceiling-bound, so the falsifier (d^{out} up) is better powered than retrenchment (share up, d^{out} down).
- **Heterogeneity by destination reach.** Out-of-block exploration lives in wide-reach (not high-volume) firms (§3) and in particular blocks (§4); let effects vary by reach and track the product and geographic margins jointly.
- **Control for intermediaries with coherence.** Low coherence flags wholesalers with spurious d^{out} ; condition on C_f or trim the low-coherence tail.
- **Carry the sample caveats.** Balanced core only (19% of firms, 62% of value); block *count* is resolution/algorithm-dependent, so fix the partition choice (BRIM, $\gamma = 1$).

Appendix: community-detection algorithms and the resolution

γ

All three algorithms partition the *same* bipartite firm–product graph, but they optimise different objectives, and the difference is entirely in the *null model* — the random baseline against which observed within-block links are scored.

Modularity and the null. Modularity rewards a partition for the links that fall inside blocks *beyond* what a degree-preserving random graph would put there. For a one-mode graph (Newman, 2006), $Q = \frac{1}{2m} \sum_{ij} (A_{ij} - \gamma \frac{k_i k_j}{2m}) \delta(c_i, c_j)$, where $k_i k_j / 2m$ is the configuration-model expectation and γ the resolution.

Louvain (Blondel et al., 2008). A fast greedy optimiser of the one-mode Q . Run on a bipartite graph it still uses the *one-mode* null, which expects firm–firm and product–product links that cannot exist; the mis-specified baseline inflates apparent within-block density and Louvain tends to *over-segment* (here: 13 blocks).

Leiden (Traag et al., 2019). An improved optimiser that repairs Louvain’s badly-connected-community defect and guarantees well-connected blocks. We run it on the same one-mode RB-modularity: a better optimiser, but the same null mis-specified for a two-mode graph (here: 9 blocks).

BRIM (Barber, 2007). Uses the *correct bipartite null*. The expected link between firm f and product p is $k_f d_p / m$ — it preserves the firm and product degree sequences separately and places expected links only across the two node types, never within one. The bipartite modularity is $Q_B = \frac{1}{m} \sum_{fp} (B_{fp} - \gamma \frac{k_f d_p}{m}) \delta(g_f, g_p)$, and blocks are co-clusters containing both firms and products. This is De Stefano’s choice and ours (here: 5 blocks, $Q_B = 0.505$). We implement it through `leidenalg`’s CPM-bipartite recipe, scaled so that $\gamma = 1$ reproduces standard Barber modularity.

The resolution γ . γ multiplies the null term — the “charge” for grouping two nodes — so $\gamma = 1$ is standard modularity, $\gamma > 1$ gives more and smaller blocks, $\gamma < 1$ fewer and larger. In the CPM form γ is a density threshold (keep a community only if its internal density exceeds γ), which is why CPM avoids the modularity resolution limit. Leiden is only the optimiser; γ belongs to the objective. We keep $\gamma = 1$ as the baseline and report $\gamma = 2$ in §4.1.

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